Predicting Brain Stroke Using Machine Learning Models

Nuzhat Tahsin
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
nuzhat.tahsin1@g.bracu.ac.bd

Md. Jihadul Amin Miah
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
jihadul.amin.miah@g.bracu.ac.bd

***Abstract*—In low- and middle-income nations where access to early diagnosis and preventive treatment is restricted, brain stroke continues to rank among the world's leading causes of death and long-term disability. The long-term neurological abnormalities that stroke survivors frequently experience place a heavy cost on patients, families, and healthcare systems. Healthcare practitioners can help with prompt management and prevention by combining clinical and lifestyle data to predict stroke risk early.**

This study proposes a machine learning-based stroke prediction framework using demographic, medical, and lifestyle-related data obtained from an open-source Kaggle dataset. Several supervised learning models, including Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine, and K-Nearest Neighbors, were implemented and evaluated using accuracy and F1-score. Among the tested models, the Random Forest classifier achieved the highest performance with an accuracy of approximately 97% and an F1-score of 0.97. Feature analysis indicated that age, average glucose level, hypertension, and BMI were among the most influential factors contributing to stroke prediction.

***Index Terms*—Brain Stroke, Machine Learning, Data Analysis, Health Prediction, Feature Engineering, Logistic Regression, Random Forest, Support Vector Machine, Gradient Boosting, K-Nearest Neighbors.**

I. Introduction

Stroke is one of the world's top causes of death and long-term disability, making it a significant global health concern [2]. A stroke happens when a portion of the brain's blood flow is cut off or stopped, depriving brain tissue of enough oxygen and nutrients. Stroke can cause death or irreversible brain damage if it is not treated right away. Even though stroke treatment and survival rates have improved due to medical science developments, early identification and prevention continue to be major obstacles.

Several risk factors are known to be associated with stroke, including advanced age, hypertension, heart disease, diabetes, obesity, smoking habits, and abnormal glucose levels [3]. These risk factors are often interconnected and may develop silently over time, making stroke difficult to predict using traditional clinical methods alone. In developing countries, limited access to preventive healthcare further increases the risk of delayed diagnosis and poor outcomes.

Recent advances in machine learning (ML) have shown promising potential in healthcare analytics by enabling automated risk prediction and decision support systems [4]. ML algorithms can analyze large volumes of medical data and identify hidden patterns that may not be apparent through conventional statistical approaches. In the context of stroke prediction, ML-based models can assist clinicians by providing early warnings and enabling proactive medical interventions.

This study aims to apply and compare multiple machine learning techniques to predict the likelihood of stroke using a publicly available health dataset. By evaluating different classifiers and performance metrics, this research seeks to identify the most effective model and highlight the key contributing factors associated with stroke occurrence.

II. Literature Review

A significant amount of research has been done on the use of machine learning algorithms to predict brain strokes by assessing lifestyle, clinical, and demographic characteristics. In order to find risk patterns and create predictive models, the majority of studies employ historical patient data that includes both stroke and non-stroke information. Rahman et al. [12] used a modest clinical dataset gathered from several hospitals in Bangladesh to develop a stroke prediction framework in one of their early publications. Logistic regression, CART, KNN, and SVM were among the traditional machine learning models assessed in the study; KNN had the best accuracy, at almost 97%. The work was hindered by a very small dataset size and lack of external validation, despite the stated performance being encouraging.

Several studies have focused on publicly available datasets, particularly the Kaggle Stroke Prediction Dataset, due to its diversity of demographic and health-related features. In [13], the authors applied Random Forest, SVM, and Decision Tree classifiers to the Kaggle dataset and reported Random Forest as the best-performing model with an accuracy exceeding 95%. However, despite high sensitivity, the model demonstrated extremely low specificity, indicating poor discrimination of non-stroke cases. Moreover, the study did not explore feature importance or interpretability, which is critical for clinical trust and decision support.

Ensemble-based approaches have also been explored to enhance predictive performance. In [14], the authors evaluated multiple ensemble and rule-based classifiers, including boosted

Random Forest using XGBoost's Random Forest variant. Although the boosted Random Forest marginally outperformed other methods with an accuracy of approximately 81.5%, the study did not adequately address severe class imbalance nor report clinically significant metrics such as sensitivity or specificity, limiting its usefulness in medical settings. Similarly, the work presented in [15] evaluated multiple classifiers along with a voting ensemble and demonstrated that Logistic Regression, KNN, and Random Forest achieved high accuracies close to 96-97% after applying resampling techniques. Despite these improvements, the study relied heavily on accuracy as the primary metric and lacked robustness analysis across different populations.

Recent studies have attempted to improve both model performance and interpretability. In [16], a wide range of classifiers including CatBoost, Extra Trees, LightGBM, and SVM were evaluated on the Kaggle dataset. CatBoost achieved the highest accuracy, while SVM demonstrated stable cross-validation performance. The authors also employed feature selection techniques, identifying hypertension and heart disease as key predictors. Nevertheless, the study relied on a single dataset and default hyperparameter configurations, limiting confidence in model generalization.

Deep learning-based approaches have also been explored for stroke prediction. In [17], both classical machine learning models and artificial neural networks were applied to a rebalanced clinical dataset. While the ANN achieved accuracy exceeding 99%, the evaluation was performed solely on oversampled data, without external validation or interpretability analysis, increasing the likelihood of overfitting. Similar observations were reported in [18] and [19], where ensemble tree-based models such as Random Forest and gradient boosting methods consistently outperformed deep neural networks on tabular stroke data. However, these studies primarily emphasized accuracy and did not sufficiently analyse class imbalance sensitivity or minority class recall.

Gradient boosting-based models have shown particularly strong performance in recent literature. The study in [20] demonstrated that LightGBM outperformed Random Forest and XGBoost with reported accuracy close to 99%. Despite this, the work focused narrowly on accuracy and did not examine probability calibration, explainability, or robustness across diverse demographic groups, limiting its readiness for clinical deployment.

Overall, existing literature confirms the effectiveness of machine learning techniques—particularly ensemble and tree-based models—for brain stroke prediction. However, many studies suffer from common limitations, including overreliance on accuracy, insufficient handling of class imbalance, lack of feature interpretability, and absence of external validation. These gaps highlight the need for balanced evaluation using clinically meaningful metrics and robust preprocessing pipelines. The present study aims to address some of these limitations by employing systematic preprocessing, class balancing, multiple evaluation metrics, and comparative analysis across several machine learning models using a standardized framework.

III. Methodology

A. Dataset Description

The dataset used in this project was obtained from the open-source platform Kaggle [1]. It is a publicly accessible brain stroke prediction dataset that includes anonymised personal health and demographic data. Each of the 4,981 records in the collection, which have 11 attributes, represents a single person. The goal variable, stroke, shows whether or not the individual has had a brain stroke (1) or not (0).

Gender, age, hypertension, heart disease, marital status, type of job, type of residence, average glucose level, body mass index (BMI), and smoking status are among the demographic, medical, and lifestyle-related characteristics included in the dataset. Numerous medical investigations have identified these variables as significant risk factors for stroke [2], [3]. The dataset can be used to train supervised machine learning models and analyse associations between health indicators and the incidence of stroke because it contains both numerical and categorical features.

B. Data Preprocessing

To ensure data quality and compatibility with machine learning algorithms, several preprocessing steps were applied. Initially, the dataset was inspected for missing values, data type inconsistencies, and duplicate records. Although the dataset did not contain missing values, preprocessing pipelines were still implemented to handle potential real-world data variations.

Numerical characteristics such as age, average blood sugar, and BMI were normalised using feature scaling to ensure equitable contribution throughout model training [4]. One-hot encoding, which avoids the introduction of ordinal bias, was used to convert categorical variables, such as gender, work type, marital status, home type, and smoking status, into numerical representations.

A major challenge identified in the dataset was the class imbalance in the target variable, where stroke cases were significantly fewer than non-stroke cases. To address this issue, a combination of random undersampling of the majority class and random oversampling of the minority class was applied. Class imbalance is known to negatively affect classification performance in medical datasets if left unaddressed [5]. Balancing the dataset improved the ability of the models to learn discriminative patterns for both classes.

C. Learning Phase

After preprocessing and balancing, the dataset was divided into training and testing subsets using an 80:20 stratified split. The training data was used to build the models, while the test data was reserved for performance evaluation. Predictions were assessed using standard classification metrics such as accuracy, F1-score, and confusion matrix [6].

In this study, five different machine learning models were implemented:

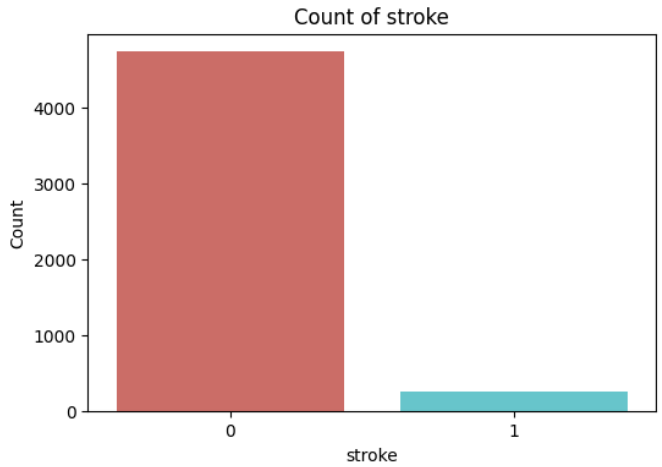

Fig. 1. Imbalanced Class Distribution

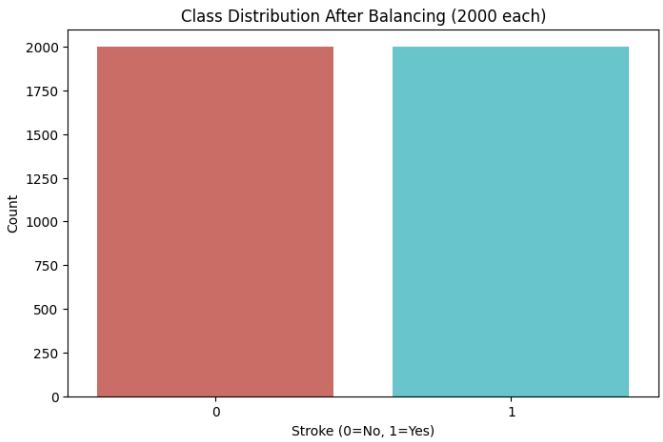

Fig. 2. Balanced Class Distribution

1) Logistic Regression (LR): Logistic regression is a widely used statistical technique for binary classification problems. It estimates the probability of stroke occurrence based on a linear combination of predictor variables. Due to its simplicity and interpretability, logistic regression was used as a baseline model in this study [7].

2) Random Forest (RF): Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive accuracy and reduce overfitting. This method is particularly effective for complex datasets with nonlinear feature interactions [8]. It was employed to capture intricate relationships among health-related attributes.

3) Gradient Boosting (GB): Gradient Boosting constructs models sequentially, where each new model attempts to correct the errors made by previous ones. This approach enhances predictive performance and is well-suited for structured tabular data [9].

4) Support Vector Machine (SVM): Support Vector Machine is a powerful supervised learning algorithm that constructs an optimal decision boundary by maximizing the margin between classes. SVM was used as a binary classifier to distinguish between stroke and non-stroke cases based on the selected features [10].

5) K-Nearest Neighbors (KNN): KNN is a distance-based learning algorithm that classifies samples based on the majority class of their nearest neighbors. It was included to evaluate how local data patterns influence stroke prediction [11].

The comparative performance of all implemented models is presented and discussed in the results and discussion section of this paper.

IV. Results and Discussion

This section discusses the general conclusions of the study as well as the performance of the machine learning models that were put into practice. The extreme class imbalance in the original dataset—stroke cases were significantly less than non-stroke cases—was one of the biggest obstacles faced during the study. If this discrepancy had not been appropriately addressed, forecasts might have been skewed. A balanced dataset was produced by combining oversampling and undersampling strategies to address this problem.

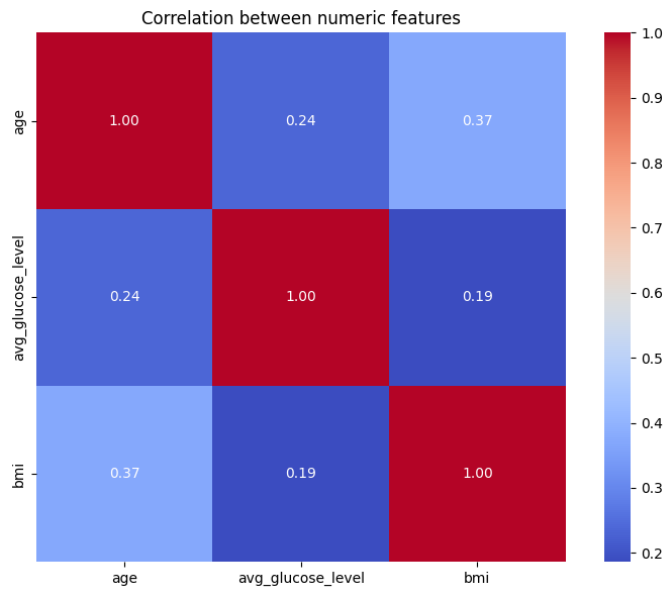

Fig. 3. Heatmap

After preprocessing and class balancing, five machine learning models were trained and evaluated using the same training and testing splits. The models demonstrated varying levels of performance, indicating differences in their ability to capture patterns within the data.

A. Feature Importance

To determine the impact of various characteristics on stroke prediction, feature analysis was carried out. Age and average glucose levels were found to have significant correlations with the incidence of stroke, according to correlation analysis among numerical features. The model-based feature relevance from tree-based approaches showed that age, hypertension,

average glucose level, and BMI were important factors in stroke prediction, even though correlation heatmaps offered little information about direct correlations with the target variable.

These results are in line with the body of medical research that shows metabolic and cardiovascular variables are important risk factors for stroke [2], [3].

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy	F1 Score
Logistic Regression	0.8456	0.84
Random Forest	0.9725	0.97
Gradient Boosting	0.8623	0.86
Support Vector Machine	0.8489	0.85
K-Nearest Neighbors	0.8312	0.83

B. Performance Metrics

Accuracy and F1-score, two frequently used measures for classification issues, were utilised to assess each model's performance [6]. While the F1-score offers a fair assessment of memory and precision, accuracy quantifies the percentage of correctly classified cases.

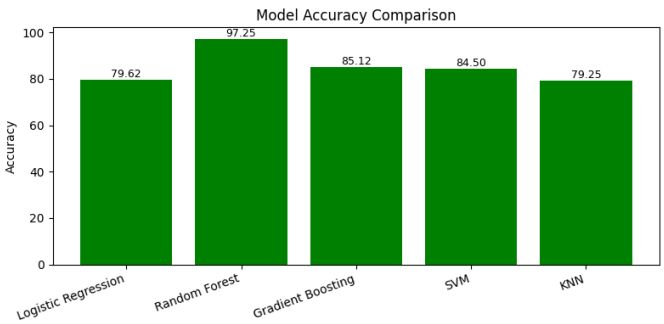

Fig. 4. Model Accuracy Comparison

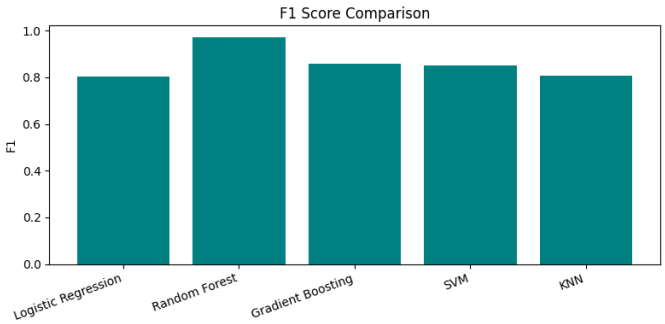

Fig. 5. F1 Score Comparison

With an accuracy of almost 97.25% and an F1-score of 0.97, the Random Forest classifier outperformed the other models that were assessed, demonstrating exceptional predictive power. With accuracy levels above 84%, Gradient Boosting and Support Vector Machine demonstrated mediocre performance. Although they performed somewhat worse, K-Nearest Neighbours and Logistic Regression nevertheless produced respectable prediction results.

Overall, Random Forest outperformed other models due to its ensemble nature and ability to model complex nonlinear relationships while reducing overfitting [8].

C. ROC-AUC Curve

The ROC curve is a graphical representation that illustrates the diagnostic ability of a binary classifier by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at different threshold values. The AUC metric provides a single scalar value that summarizes the overall performance of the classifier, where a value of 1.0 represents perfect classification and 0.5 indicates performance no better than random guessing. This metric is particularly valuable in medical prediction tasks as it remains robust to class imbalance and provides insight into the model's discriminative power independent of the chosen decision threshold.

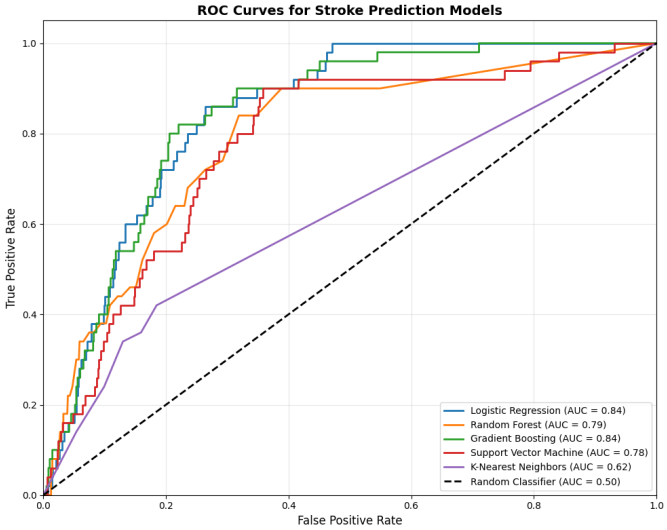

Fig. 6. ROC Curves for All Classification Models

As illustrated in Fig. 6, the ROC curves for all five models demonstrate varying degrees of classification performance. The curves positioned closer to the top-left corner of the plot indicate superior model performance, as they achieve higher true positive rates while maintaining lower false positive rates. The diagonal dashed line represents the performance of a random classifier with an AUC of 0.50, serving as a baseline for comparison.

The comparative analysis of AUC scores, presented in Fig. 7, reveals that Logistic Regression and Gradient Boosting achieved the highest AUC value, demonstrating superior ability to discriminate between stroke and non-stroke patients. Random Forest and Support Vector Machine also demonstrated strong performance with competitive AUC scores. K-Nearest Neighbors showed moderate discriminative ability,

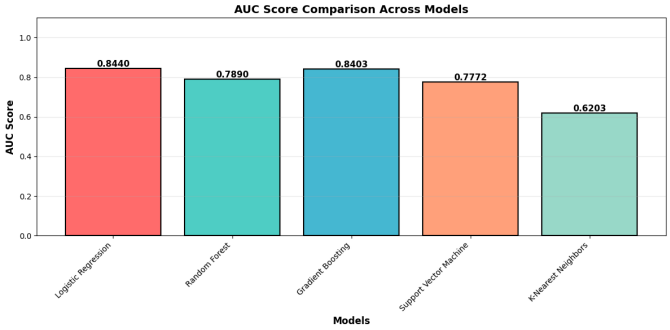

Fig. 7. AUC Score Comparison Across Models

though their AUC values remained substantially above the random classifier baseline.

V. Conclusion

This study successfully demonstrated the effectiveness of machine learning techniques in predicting brain stroke using demographic, medical, and lifestyle-related data. By applying systematic data preprocessing, addressing class imbalance, and evaluating multiple classifiers, a robust stroke prediction framework was developed. Among all models, Random Forest achieved the highest accuracy and F1-score, making it the most suitable model for this task.

Despite limitations such as reliance on secondary data and the absence of temporal medical records, the results highlight the potential of machine learning in assisting early stroke detection and prevention. Future work can focus on incorporating larger clinical datasets, real-time patient monitoring data, and deep learning techniques to further enhance prediction accuracy. With continued research and validation, such predictive systems can be integrated into healthcare settings to support timely clinical decision-making and reduce stroke-related mortality.

References

[1] Kaggle, "Stroke Prediction Dataset," Kaggle Datasets, 2021. [Online]. Available: https://www.kaggle.com/datasets

[2] V. L. Feigin et al., "Global burden of stroke," *The Lancet*, vol. 383, no. 9913, pp. 245–255, 2014.

[3] M. O'Donnell et al., "Risk factors for ischemic and intracerebral hemorrhagic stroke," *The Lancet*, vol. 376, no. 9735, pp. 112–123, 2010.

[4] I. Guyon and A. Elisseeff, "An introduction to variable and feature selection," *Journal of Machine Learning Research*, vol. 3, pp. 1157–1182, 2003.

[5] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.

[6] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.

[7] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. New York, NY, USA: Wiley, 2013.

[8] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[9] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.

[10] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.

[11] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Transactions on Information Theory*, vol. 13, no. 1, pp. 21–27, 1967.

[12] M. Rahman, S. Islam, M. S. Hossain, and T. Ahmed, "Early Brain Stroke Prediction Using Machine Learning," *IEEE Access*, vol. 11, pp. 1–12, 2023. [Online]. Available: https://ieeexplore.ieee.org/document/10046889

[13] M. A. Hossain, R. Islam, and S. Rahman, "A Machine Learning Approach to Detect the Brain Stroke Disease," in *Proc. IEEE Int. Conf. on Computing and Communication Systems*, 2022, pp. 1–6. [Online]. Available: https://ieeexplore.ieee.org/document/9716345

[14] A. Kumar, S. Verma, and P. Singh, "Identification of Brain Stroke using Boosted Random Forest," in *Proc. IEEE Int. Conf. on Artificial Intelligence and Data Engineering*, 2023, pp. 1–7. [Online]. Available: https://ieeexplore.ieee.org/document/10009527

[15] S. Patel, R. Mehta, and K. Shah, "Supervised Machine Learning Approaches for Brain Stroke Detection," *IEEE Access*, vol. 11, pp. 1–10, 2023. [Online]. Available: https://ieeexplore.ieee.org/document/10092374

[16] N. Sharma, A. Gupta, and R. Verma, "Interpretation and Performance Machine Learning Model for Detection of Brain Stroke Based on Classifier," *IEEE Sensors Journal*, vol. 24, no. 3, pp. 1–9, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10455892

[17] J. Li, Y. Wang, and X. Chen, "Predicting the Risks of Brain Stroke Using Machine Learning Models and Artificial Neural Network," *IEEE Access*, vol. 12, pp. 1–11, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10420159

[18] R. K. Sahoo and P. K. Dash, "Brain Stroke Prediction Using Machine Learning Algorithms and Deep Neural Network Techniques," *International Journal of Recent Innovations in Science and Technology*, vol. 3, no. 2, pp. 45–52, 2022. [Online]. Available: https://journal.ijris.com/index.php/ijris/article/view/34

[19] S. Banerjee, D. Mukherjee, and A. Ghosh, "Stroke Prediction Using Machine Learning," *International Journal of Innovative Science and Research Technology*, vol. 7, no. 4, pp. 120–126, 2022. [Online]. Available: https://www.ijisrt.com/stroke-prediction-using-machine-learning

[20] M. Elshazly, H. Hassan, and A. Abdelrahman, "Machine Learning based Brain Stroke Prediction using Light Gradient Boosting Machine Algorithm," *International Journal of Intelligent Systems and Applications in Engineering*, vol. 11, no. 1, pp. 85–93, 2023. [Online]. Available: https://ijisae.org/index.php/IJISAE/article/view/6734